

write the 10 sentences with process of this

① Sentence

1 I love playing cricket.

* Converting in the tokens (from the tokenization)
["I" "love" "playing" "cricket"]

* Converting the Embedding

I = [0.18, 0.45, 0.67]

love → [0.90, 0.30, 0.34]

playing → [0.30, 0.77, 0.11]

cricket → [0.90, 0.30, 0.93]

② Itaninada is a smart city

* Converting the tokens (from the tokenization)
["Itaninada" "is" "a" "smart" "city"]

* Converting the embedding

Itaninada → [0.12, 0.13] city [0.5, 0.10, 0.11]

is → [0.4, 0.6]

a →

smart → [0.20, 0.31]

Signature: _____

③ He bought a new laptop

Convert tokens (from precess tokenization)

["He" "bought" "a" "new" "laptop"]

Convert the in to Embeddings

He - $[0.6, 0.4, 0.2]$

bought - $[0.98, 0.21, 0.92]$

a - $[0.1, 0.2, 0.3]$

new - $[0.21, 0.32, 0.41]$

Laptop - $[0.46, 0.53, 0.62]$

The cat is sleeping

Convert tokens (from precess tokenization)

["The" "cat" "is" "sleeping"]

Convert the in to the embeddings for precess

The $\rightarrow [0.21, 0.21, 0.18]$

cat $\rightarrow [0.41, 0.52, 0.55]$

is $\rightarrow [0.11, 0.10, 0.2]$

sleeping $\rightarrow [0.52, 0.62, 0.65]$

The food tastes delicious

Convert tokens (from precess tokenization)

["The" "food" "tastes" "delicious"]

Convert the precess to Embeddings

The $\rightarrow (0.24, 0.31, 0.5)$
Good $\rightarrow (0.15, 0.25, 0.35)$
Tastes $\rightarrow (0.17, 0.27, 0.37)$
Delicious $\rightarrow (0.31, 0.32, 0.31)$

I am learning machine learning

Convert token (in process of tokenization)
["I" "am" "learning" "machine" "learning"]

Convert the form of embedding terms

I $\rightarrow (0.4, 0.3, 0.2)$
am $\rightarrow (0.27, 0.28, 0.31)$
learning $\rightarrow (0.71, 0.73, 0.74)$
machine $\rightarrow (0.81, 0.82, 0.83)$
learning $\rightarrow (0.91, 0.94, 0.95)$

The sun rises in the east

Converting tokenization

["the" "sun" "rises" "in" "the" "east"]

Embedding

the $[0.12, 0.45, 0.51]$
sun $[0.92, 0.22, 0.55]$
rises $[0.34, 0.88, 0.71]$
east $[0.26, 0.19, 0.84]$

plain text

sun $\rightarrow [0.92, 0.22, 0.55] \rightarrow$ Transformer processing
"plain text"

"rise" pays attention to "sun"

"east" pays attention to "rise"

current token

rises

east

sun

Transformer

sun $\rightarrow$ tokens $\rightarrow$ rises
direction $\rightarrow$ east

pays attention to

sun (80%)

rises (70%)

rises (60%)

* Next Token Prediction

Next	—	0%
east	—	95%
west	—	3%
sky	—	1%
moving	—	1%

The sun rises in the east

• Birds are flying in the sky

Tokenization

{ "Birds" "are" "flying" "in" "the" "sky" }

embedding Embeddings

Birds — { 0.12, 0.23, 0.31 }

are — { 0.31, 0.5, 0.32 }

flying — { 0.36, 0.71, 0.31 }

in — { 0.18, 0.7, 0.8 }

sky — { 0.43, 0.55, 0.53 }

Signature : _____

Plan derd € Birds [0.12, 0.23, 0.31]

Transformation

This process are attention pos of the

Birds - pos attention dlsing

dlsing pos attention on stgs

Current

Birds

flying

attention

dlsing 80%

stgs 20%

~~Nete~~ Token Prediction

Birds are dlsing in the stgs →

stgs — 96%

water — 2%

land — 1%

Boat — 1%